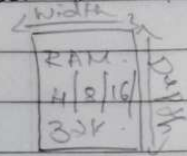


Block Memory on FPGA

Block RAM (BRAM)

- BRAM is equal to the 'Embedded memory' inside an FPGA
- It is also called Embedded Block RAM (EBR)
- Discrete resource, only a limited number per FPGA
- Amount of BRAM depends on
 - > FPGA family
 - > FPGA size & cost
- Used to store large amounts of on-chip data



BRAM Characteristics.

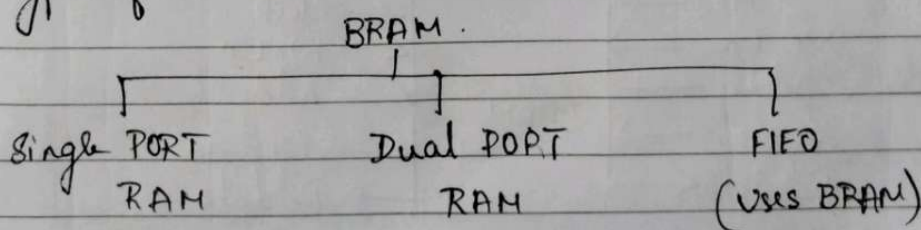
Features:

- Fixed memory block sizes
- Common sizes
 - 4Kb
 - 8Kb
 - 16Kb
 - 32Kb
- Configurable width and depth
- Operates synchronously (clock based)
- Can be initialized during FPGA programming

Applications.

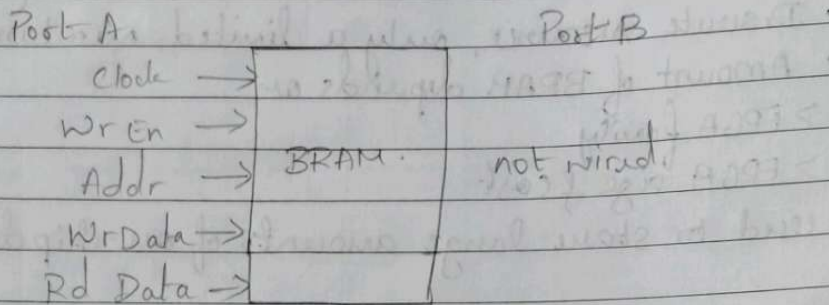
- Buffers
- Lookup tables (LUTs)
- FIFOs

Types of BRAM



Single Port BRAM

- Has only one memory port (Port A)
- Performs either Read OR write in one clock cycle
- Simplest BRAM configuration.
- Uses the least FPGA resources.

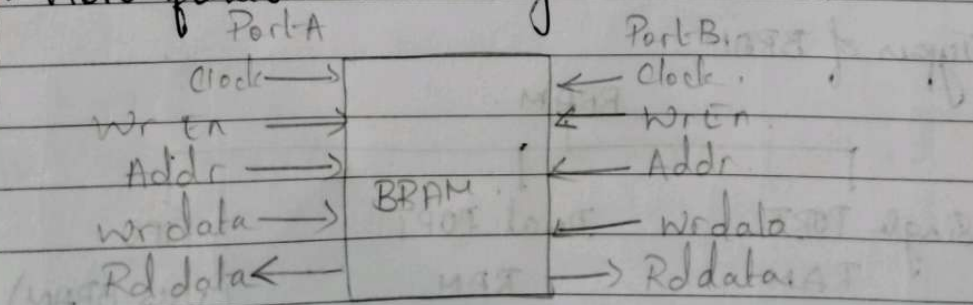


Operation

- Clk-based operation
- $Wr En = 0 \rightarrow$ Read
- $Wr En = 1 \rightarrow$ write
- Only one address accessed per clock cycle.
- Cannot read and write simultaneously.

Dual - Port BRAM (DPRAM)

- Has two independent ports.
 - Port A
 - Port B
- Each port can
 - Read
 - Write.
- Both ports work simultaneously.
- More flexible than single - Port BRAM.



operation.

- Both ports are clocked
- Can access:
 - Different addresses
 - During the same clock cycle

eg:

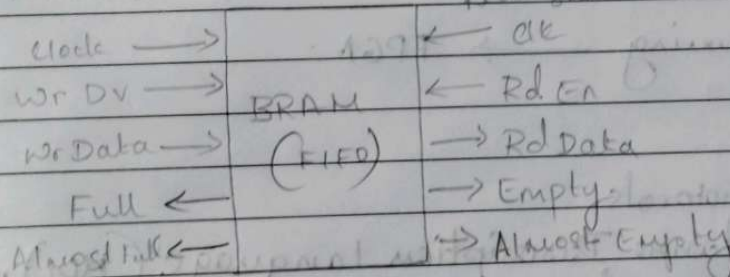
- Port A → Read Address 0
- Port B → write Address 200
- Both occurs in the same cycle.

FIFO using BRAM.

- FIFO is First In First out.
- Data is read in the same order it is written.
- Internally built using Dual-Port BRAM.
- Commonly used in FPGA design.

Write Side

Read Side



ASIC Design Flow

Design specification

↓
RTL coding

↓
Functional Simulation

↓
Coverage Analysis

↓
Logic Synthesis

↓
DFT Synthesis

↓
ATPG Vector Generation

Logic Equivalency checking

↓
Floor Planning

↓
Power Planning

↓
Placement

↓
CTS

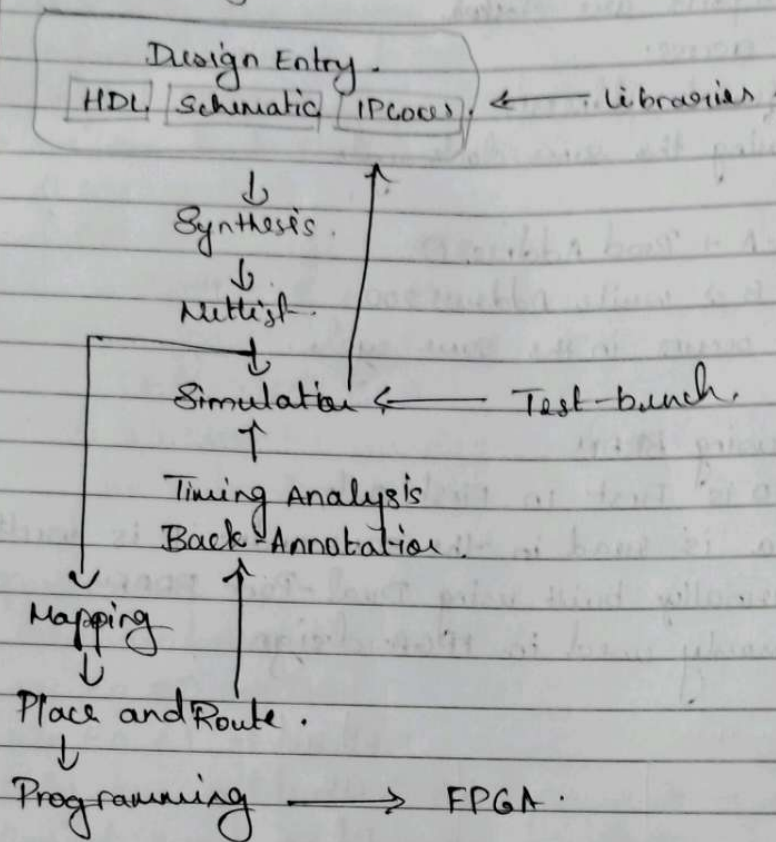
↓
Routing

↓
Static Timing Analysis

↓
Gate level Simulation

↓
GDS II

FPGA Design Flow.

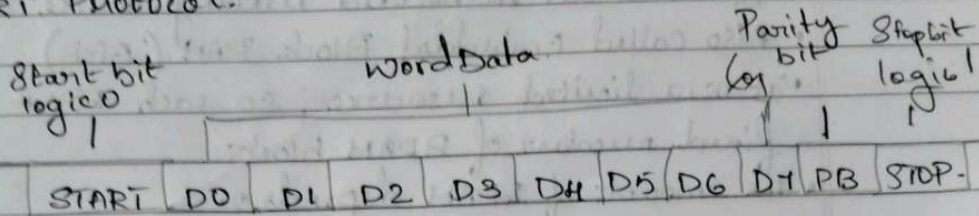


VART Protocols.

- A Hardware Description Language (HDL) used for modeling digital system.
- Allows designers to describe the behavior and structure of electronic systems.
- Verilog AMS used for modeling Analog & Mixed signal language.
- Origin.
 - Developed in 1984 by Gateway Design Automation.
 - Standardized as IEEE 1364 in 1995.
- Versions.
 - Verilog - 1995: The first IEEE standard.
 - Verilog - 2001: Improved features and usability.
 - Verilog - 2005: Minor updates & clarifications.

- System Verilog (2005): Extension to Verilog for design and verification.

UART Protocol.



Start by detecting transition from logic 1 to logic 0

Incoming data sampled at the bit-pulse center

Sample stop bits